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Sarosh Khan, MBBS, Uzma Sajjad, FCPS, Samer Fawaz, MD, Haroun Butt, MBChB, Rupert Simpson, MBChB, Abdalla Ibrahim, MBBS, Craig Robertson, BSc, Paul Kelly, MD, Shah R. Mohdnazri, MD, Kare Tang, BM, Christopher M. Cook, PhD, Ozan M. Demir, PhD, Peter O'Kane, MD, James C. Spratt, MD, Emmanouil S. Brilakis, PhD, Grigoris V. Karamasis, MD, Matthew Shun-Shin, PhD, Rasha Al-Lamee, PhD, Thomas R. Keeble, MD, John R. Davies, PhD

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A Randomized, Placebo-Controlled Trial of Chronic Total Occlusion Percutaneous Coronary Intervention in Stable Angina - ORBITA-CTO

Brief Title: Randomized, Placebo-Controlled Trial of CTO PCI

Sarosh Khan MBBS^{a,b}, Uzma Sajjad FCPS^{a,b}, Samer Fawaz MD^a, Haroun Butt MBChB^{a,b}, Rupert Simpson MBChB^{a,b}, Abdalla Ibrahim MBBS^{a,b}, Craig Robertson BSc^a, Paul Kelly MD^a, Shah R. Mohdnazri MD^a, Kare Tang BM^a, Christopher M. Cook PhD^{a,b}, Ozan M. Demir PhD^{a,b}, Peter O’Kane MD^c, James C. Spratt MD^d, Emmanouil S. Brilakis PhD^e, Grigoris V. Karamasis MD^v, Matthew Shun-Shin PhD^g, Rasha Al-Lamee PhD^g, Thomas R. Keeble MD^{a,b} and John R. Davies PhD^{a,b*}

^a Department of Interventional Cardiology, Essex Cardiothoracic Centre, Mid and South Essex NHS Foundation Trust, Essex, United Kingdom, ^b Anglia Ruskin University School of Medicine and MTRC, Chelmsford, Essex, United Kingdom, ^c Department of Interventional Cardiology, Royal Bournemouth Hospital, Bournemouth, United Kingdom, ^d Department of Interventional Cardiology, St. George’s Hospital, London, United Kingdom, ^e Minneapolis Heart Institute and Minneapolis Heart Institute Foundation, Abbott Northwestern Hospital, Minneapolis, United States, ^f School of Medicine, Attikon University Hospital, National and Kapodistrian University of Athens, Athens, Greece, ^g National Heart and Lung Institute, Imperial College London, London, United Kingdom

*Address for correspondence:

Dr John Davies, Department of Interventional Cardiology, Research office, Level B, Essex Cardiothoracic Centre, Essex, United Kingdom, SS16 5NL.

Email: john.davies32@nhs.net

Telephone: +44 7961 424133

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Abstract

Background

Percutaneous coronary intervention for coronary chronic total occlusion (CTO PCI) is offered for symptom and quality of life improvement, despite the absence of blinded randomized evidence.

Objectives

The aim of this study was to assess the efficacy of CTO PCI in the first randomized, placebo-controlled trial of CTO PCI.

Methods

ORBITA-CTO was a multi-center randomized, blinded trial comparing CTO PCI with a placebo procedure. Patients had angina attributable to a single-vessel CTO, without bystander coronary disease. Angina symptoms were recorded daily using the ORBITA-app. After dual-injection coronary angiography, patients were randomized to either CTO PCI or placebo. Blinding was maintained using auditory isolation and deep conscious sedation. Anti-anginal medications were stopped at randomization and re-introduced on a patient-initiated protocol. At 6-month follow-up, assessments were repeated. The primary efficacy outcome was the angina symptom score, an ordinal scale combining the daily symptom burden assessed by ORBITA-app, anti-anginal use, and override events. Secondary outcomes included symptom and quality-of-life questionnaires and blinding fidelity. Trial registered at Clinicaltrials.gov (identifier: NCT05142215).

Results

Between Oct 19, 2021, and Oct 21, 2025, 50 patients were randomly assigned to CTO PCI (n=25) or placebo (n=25). One patient randomized to PCI was withdrawn during the procedure due to a complication. All 50 patients were included in the primary analysis. Compared to placebo, CTO PCI resulted in an immediate and sustained improvement in angina symptom

score (OR 4·38, 95% CrI 1·57 to 12·69, Pr[Benefit]=0.996), arising from a clear reduction in the number of episodes of angina (OR 4·38, 95% CrI 1·55 to 11·78, Pr[Benefit]=0.997). This resulted in an additional 30·6 days (95% CrI 11·1 to 50·7, Pr[Benefit]>0.999) free of angina. Improvements were also observed in SAQ angina frequency (+10·7, 95% CrI 1·4 to 20·2, Pr[Benefit]=0.988), physical limitation, quality of life, summary score, and CCS class. Blinding of patients, staff and researchers was maintained.

Conclusions

In patients with symptomatic single-vessel CTO, CTO PCI improves angina beyond placebo.

Condensed Abstract

The symptomatic benefit of percutaneous coronary intervention (PCI) for chronic total occlusion (CTO) remained uncertain due to the absence of blinded randomized evidence. ORBITA-CTO was a multicenter, randomized, placebo-controlled trial enrolling patients with angina and single-vessel CTO. Patients were assigned to CTO PCI or placebo with full blinding. CTO PCI produced an immediate and sustained improvement in angina symptom score compared with placebo, driven primarily by reduced angina frequency, with parallel improvements in quality-of-life measures. For symptomatic single-vessel CTO, PCI improves angina beyond placebo.

Key words

Chronic total occlusion, chronic coronary syndromes, percutaneous coronary intervention, placebo-controlled

Abbreviations list

CTO Chronic total occlusion

PCI Percutaneous coronary intervention

MDT multi-disciplinary team

CCS Canadian Cardiovascular Society

NYHA New York Heart Association

LAD left anterior descending artery

LCx left circumflex artery

RCA right coronary artery

Clinical trial: NCT05142215

Introduction

Percutaneous coronary intervention to coronary artery chronic total occlusions (CTO PCI) is routinely undertaken to relieve angina and improve quality of life, despite the absence of blinded randomized evidence demonstrating benefit beyond placebo.^{1,2} Existing evidence for symptomatic improvement is derived largely from observational studies and open-label randomized trials,^{3,4} with the two largest randomized studies reporting conflicting results and contributing to uncertainty reflected in contemporary guideline recommendations.^{5,6} Consequently, uncertainty remains regarding the extent to which symptom relief reflects true physiological improvement rather than a placebo effect.

The ORBITA (Objective Randomized Blinded Investigation with Optimal Medical Therapy of Angioplasty in Stable Angina) trial demonstrated a substantial placebo effect associated with contemporary non-CTO PCI in patients with chronic coronary syndromes, fundamentally altering the interpretation of symptom-based outcomes after coronary intervention and emphasized the need for placebo control in trials testing the efficacy of interventional procedures aimed at symptom relief.⁷ Given the greater technical complexity, longer procedural duration, and higher resource intensity of CTO PCI, coupled with lower success rates and higher complication rates than conventional non-CTO PCI, testing if there is a physiological treatment effect of CTO PCI beyond placebo is particularly important.^{8–10}

The ORBITA-CTO study is the first double-blind, placebo-controlled, randomized trial to evaluate the effect of CTO PCI on angina and quality of life in patients with stable angina receiving background anti-anginal therapy.

Methods

Study design and patients

ORBITA-CTO was an investigator initiated, randomized, double-blind, placebo-controlled trial conducted at two hospitals in the UK (Appendix, p6). The trial protocol has been published¹¹ and is available in the Appendix (p94). The study was approved by the Yorkshire & The Humber - Bradford Leeds Research Ethics Committee (reference: 21/YH/0165). The trial conduct was overseen by an independent data safety and monitoring board (DSMB) who also adjudicated all study adverse events. Authors SK and JRD attest to the accuracy and completeness of the data and adherence to the protocol. All authors contributed to the writing and approval of the final version submitted for publication.

Patients were eligible to be enrolled if they had angina, a single-vessel CTO, defined as a coronary artery occlusion of at least 3 months' duration or a probable CTO (Thrombolysis in Myocardial Infarction grade 0 flow through the lesion with no thrombus, no staining at the proximal cap, and presence of mature collaterals)¹² when duration was unknown. The CTO vessel was required to be at least 2.5 mm in diameter, with no angiographically significant disease in non-CTO vessels ($\geq 50\%$ left main stem or $\geq 70\%$ stenosis in other epicardial vessels ≥ 2 mm). Evidence of inducible ischemia and viability within the CTO-subtended myocardial territory was required. Additional inclusion criteria were acceptance for CTO PCI by a specialist CTO operator and a Japanese CTO (J-CTO) score of ≤ 3 . Patients were screened for suitability by the regional CTO and complex PCI multidisciplinary team (MDT) at the primary site. J-CTO scores were determined by MDT consensus by experienced CTO operators and reconfirmed by operators after dual-access coronary angiography at the randomization procedure. The MDT met monthly and included at least two CTO operators, a general cardiologist, a cardiothoracic surgeon, a cardiac radiologist, specialist PCI nurses, and research staff. All patients underwent a structured telephone assessment by senior PCI nurses

before MDT discussion, and patients were invited to join MDT meetings by video consultation when clarification of symptoms was required. A complete screening log was maintained for all CTO cases discussed at the primary site during the recruitment period (Appendix, p72). Patients enrolled from alternate sites did not have a formal screening log, but cases were reviewed with ORBITA-CTO investigators and are included in the log.

Key exclusion criteria were: age younger than 18 years, recent acute coronary syndrome (<4 weeks), PCI to a non-CTO lesion within 4 weeks, clinically important non-CTO coronary disease requiring revascularization, ischemia in a non-CTO territory, contraindications to PCI or drug-eluting stent implantation, inability to tolerate dual antiplatelet therapy; the complete exclusion criteria are available in the Appendix (p10). Biological sex was self-reported by patients.

At enrolment, eligibility was reconfirmed and written informed consent obtained. Patients completed the Seattle Angina Questionnaire (SAQ) and EQ-5D-5L, including the visual analogue scale. Investigators assessed angina severity using the Canadian Cardiovascular Society (CCS) classification and breathlessness using the New York Heart Association (NYHA) classification and the Rose Dyspnea Scale (RDS).

Patients were trained to use a dedicated smartphone application (ORBITA-app) to record daily angina episodes throughout the trial.^{13,14} If data entry was missed on any day, patients were contacted by a blinded research fellow to ensure completion.

Patients then underwent protocol-directed optimization of anti-anginal therapy to a minimum of three anti-anginal units; details of the anti-anginal medication protocol are shown in the

Appendix (p17). Once the minimum therapy regimen was achieved for 48 hours, patients were required to report angina during a minimum of one week before pre-randomization assessments and randomization. Patients who were asymptomatic after optimization were withdrawn from the trial. During this period, dual antiplatelet therapy and proton-pump inhibitors were initiated or continued in all patients. Anti-anginal medication changes could only be made by the blinded research team.^{13,14}

Following the symptom assessment phase, patients attended hospital for repeat symptom and quality-of-life questionnaire assessment followed by research cardiac catheterization randomization procedure. On admission to the day-case unit, patients were advised to turn off their mobile phones and electronic devices, and all clocks in patient areas, catheterization laboratories, transit corridors and wards were removed.

Randomization and masking

Patients wore noise-cancelling headphones playing music to establish auditory isolation throughout the procedure and incremental doses of intravenous sedation and analgesia were administered to maintain a deep level of conscious sedation, where patients could maintain their own airway but were otherwise unrousable. Dual access coronary angiography was performed. The coronary angiogram was reviewed to confirm persistence of the CTO, absence of clinically significant bystander disease, and to reconfirm a J-CTO score of 3 or less; any contravention led to withdrawal from the study. Patients were randomly assigned (1:1) to receive either CTO PCI or placebo with a validated, automated, online randomization platform (Randi, open-source application) with a variable block size (between 4 and 12) algorithm with no stratification. Patients, ward staff, subsequent clinical caregivers, and blinded members of the research team were masked to treatment allocation. The operators

and research staff present during the randomization procedure had no further contact with patients after the procedure. Blinding indices for patients, ward staff and blinded research staff were assessed at discharge using validated methods.¹⁵

Procedures

In the CTO PCI group, two consultant grade specialist CTO operators used contemporary techniques, algorithms, and equipment to perform the procedure. Recanalization strategy was at the operators' discretion. Intravascular imaging was encouraged. A technically successful procedure was defined as thrombolysis in myocardial infarction (TIMI) flow 2-3 with <30% residual angiographic stenosis in the CTO segment. Invasive coronary physiological assessment using intravenous adenosine (140 µg/kg per min) to induce hyperemia was performed in the CTO PCI group after completion of revascularization.

In the placebo group, guide catheters were removed after randomization. Bilateral arterial sheaths remained in-situ. A deep level of conscious sedation was maintained for a total of 60 minutes after randomization. Operators remained scrubbed, manipulated sheaths, and simulated the PCI procedure. To simulate procedural sensations, patients received two 1-minute weight-adjusted intravenous adenosine infusions.

All patients were transferred directly from the catheterization laboratory to the cardiology ward (distinct to the day-case unit that patients attended on admission) and remained overnight. A scripted handover was used to minimize the risk of unblinding. Blinding of patients, ward staff, and research personnel was assessed before discharge the following day or later if prolonged stay was necessary. Prolonged admission was defined as a serious adverse event. All patients were discharged on dual antiplatelet therapy as clinically

indicated. Intra-procedural complications necessitating unblinding for patient safety resulted in withdrawal from the trial.

Follow-up

On the day of the randomization procedure, all anti-anginal medications were stopped.

Patients entered a 24-week blinded follow-up phase during which they continued daily symptom reporting using the ORBITA-app. During this phase, initiation and up-titration of antianginal medications were triggered by patients and were managed by blinded trial personnel, with the use of a pre-defined stepwise anti-anginal up-titration protocol.

At the end of blinded follow-up, patients completed repeat symptom assessments and questionnaires, after which point, treatment allocation was disclosed. Patients then entered an unblinded follow-up period in which they continued app-based symptom reporting for a further two weeks. Patients then returned to routine clinical care. Patients assigned to placebo who remained symptomatic after unblinding were offered CTO PCI, typically within one month.

Outcomes

The primary efficacy outcome was the angina symptom score, an ordinal daily health-status scale derived from the number of angina episodes recorded on a given day and the number of anti-anginal medication units prescribed on that day, with definitions and scoring provided in the Appendix (p12-16). The ordinal scale also incorporated prespecified escalation states, including unblinding due to unacceptable angina, acute coronary syndrome (defined as unstable angina or acute spontaneous myocardial infarction meeting the Fourth Universal

Definition of Myocardial Infarction), and death, which yielded the maximum score (Appendix, p16).

Secondary efficacy outcomes included patient-reported angina and quality-of-life measures (SAQ physical limitation, angina frequency, angina stability, treatment satisfaction and quality of life domains and SAQ summary score and EQ-5D-5L respectively) and physician-assessed symptom severity (CCS, NYHA and RDS). CCS and RDS grades range from class 0 (no angina or breathlessness except with strenuous exertion) to class IV (symptoms at rest or with minimal exertion), and NYHA class ranges from I (no limitation of physical activity) to IV (symptoms at rest).

Prespecified methodological outcomes were fidelity of blinding at discharge following the randomization procedure and protocol adherence to assigned treatment allocation. The Bang blinding index ranges from -1 to +1; +1 suggests complete unblinding, 0 suggests no patients knew their allocation and -1 indicates systematic opposite guessing. A threshold of -0.2 to 0.2 demonstrated an acceptable fidelity of blinding. Protocol adherence was defined as patient cross over from allocated arm to other (i.e. placebo to CTO PCI arm).

Sample size

At the time of trial inception, no randomized, placebo-controlled data in CTO PCI were available, and the operating characteristics and expected effect size of the ORBITA-app-derived symptom outcome were unknown. Accordingly, sample size was determined using a conservative and pragmatic approach informed by evidence from blinded and unblinded revascularization studies, and the requirement to preserve masking fidelity and procedural feasibility in a technically complex intervention. For planning purposes, a frequentist

framework was used as an approximation of the operating characteristics of the prespecified Bayesian models. Assuming a moderate-to-large treatment effect that would clinically justify the complexity of CTO PCI, and consistent with effects observed in blinded PCI trials and smaller than those reported from unblinded CTO studies, a total of 44 evaluable patients would provide 80% power at a two-sided significance level of 0.05 to detect a clinically meaningful difference in ordinal angina burden. Allowing for approximately 10% withdrawal or attrition, 50 patients were required. This sample size was also considered appropriate pragmatically, balancing the ability to detect a meaningful effect with the resource intensity of CTO PCI and the requirement for delivery by specialist operators under blinded conditions.

Statistical analysis

All analyses were conducted according to the intention-to-treat principle for primary analyses. The primary outcome was analyzed using a Markov Ordinal State-Transition (MOST) model within a Bayesian framework to exploit the daily repeated measures. This model has previously been described,¹⁶ used within other cardiovascular trials, and further details available in the Appendix (p23). In brief, a Markov ordinal transition model was constructed, in which the odds of transitioning to a lower symptom-score state on a given day were conditioned on the previous day's symptom score (first-order), baseline symptom burden during the pre-randomization assessment period, trial day number, and randomized group (CTO PCI vs placebo). Random intercepts were added. Trial day number was allowed to interact with treatment group to permit time-varying treatment effects. Non-linear effects were modelled using restricted cubic splines, and partial proportional odds constraints were applied with respect to time where appropriate. Treatment effects are reported as odds ratios (ORs), constructed such that an OR greater than 1 indicates a higher probability of

improvement. We present 95% credible intervals (CrI) and the probability of benefit ($\text{Pr}[\text{Benefit}]$). In addition to the daily OR clinically interpretable contrasts were derived from the fitted transition probabilities, including the probability of being in a state of no angina for an exemplar patient with median pre-randomization characteristics.

Prespecified measures of trial conduct were fidelity of blinding and adherence to assigned treatment allocation, defined as absence of crossover between groups during blinded follow-up. Blinding was assessed for patients, ward staff, and blinded research personnel at discharge following the randomization procedure and at follow-up using a validated blinding index with corresponding confidence intervals; full details of the blinding framework and thresholds are provided in the Appendix (p22).

Secondary outcomes were assessed at pre-randomization and at 6-month follow-up and included Seattle Angina Questionnaire domain scores and summary score, EQ-5D-5L index values and visual analogue scale scores, and physician-assessed Canadian Cardiovascular Society angina class, New York Heart Association functional class, and Rose Dyspnea Scale grade. For continuous and ordinal secondary outcomes, a Bayesian ordinal proportional odds model was used, with follow-up values conditioned on pre-randomization values modelled using restricted cubic splines and on randomized group. No treatment interactions were prespecified for secondary outcomes. For clinical interpretation, contrasts between groups are presented for a typical patient with the median pre-randomization value, transformed to the original measurement scale, with corresponding 95% credible intervals and probability of benefit. Further details, including prior specifications, are provided in the Appendix (p22-26).

EQ-5D-5L responses were converted to index values using the UK value set via the EuroQol crosswalk methodology, and distributions of responses across EQ-5D-5L domains are presented in the Appendix (p60). The ORBITA-app daily symptom diary is designed to minimize missing entries. The longitudinal ordinal modelling framework accommodates intermittent missingness under a missing-at-random assumption by conditioning on observed history and pre-randomization symptom burden.

Analyses were conducted using the statistical environment “R” (version 4.5.1). Bayesian ordinal modelling was performed using package “rmsb” (version 1.1-2),¹⁷ and blinding indices were calculated using package BI. Model output, diagnostics, and prior specifications are provided in the Appendix (p23). The trial was registered at Clinicaltrials.gov (identifier: NCT05142215).

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Between October 19, 2021, and October 21, 2025, 418 patients were assessed for eligibility, of whom 83 met the trial eligibility criteria. Fifty-six patients were enrolled (three patients were re-enrolled), and 50 proceeded to randomization to either CTO PCI (25 patients) or placebo (25 patients). One patient allocated to the CTO PCI group was withdrawn during the randomization procedure owing to a coronary perforation requiring pericardiocentesis necessitating unblinding. This was considered an override event equivalent to an acute coronary syndrome and contributes to the primary outcome. No deaths, myocardial

infarctions, or unblinding due to intractable angina, occurred in either group, and no patients crossed over from the placebo to PCI arm.

The final intention-to-treat analysis for the primary endpoint consisted of all 50 randomized patients, and for the secondary symptom endpoints comprised the 24 patients in the CTO PCI group and 25 in the placebo group with blinded follow-up data (*Figure 1*). Median follow-up was 174 days (IQR 171-181).

Pre-randomization characteristics are shown in *Table 1*. Most patients were male (37[74%] male; 13 [26%] female). The median age was 64 years (IQR 56-71) and most patients had previous PCI (29 patients [58%]). At pre-randomization, most patients were in CCS class II or III (47 [94%]), NYHA class II-III (42 [84%]) and RDS class 1-2 (40 [80%]). The CTO vessel was the right coronary artery (RCA) (30 [60%]), left anterior descending artery (LAD) (16 [32%]), and left circumflex (LCx) (4 [8%]). The median J-CTO score was 2 (IQR 1 to 2) and PROGRESS score 1 (IQR 0 to 1). The median ischemic burden was 18% (IQR 13-20), and the median left ventricular ejection fraction was 55% (IQR 54-60).

Procedural characteristics are summarized in *Table 2*. Dual arterial access was used in all procedures. Median JCTO score was 2 (IQR 1 to 2) in PCI group and 1 (IQR 0 to 2) in placebo group. Mean (SD) lesion stent length was 69.4 mm (± 29.8) In the CTO PCI group, successful crossing was achieved in 96% (24/25) with antegrade wire escalation (AWE, 68%) the predominant technique, and intravascular imaging used in 92% (23/25) of procedures. CTO PCI procedures were longer than placebo procedures (191 ± 52 vs 104 ± 14 mins) and were associated with greater fluoroscopy time, contrast volume, and radiation exposure.

At pre-randomization, median angina frequency was lower in the CTO PCI group than in the placebo group, whereas the median number of anti-anginal medication units was higher (CTO PCI 4 (IQR 3 to 5) and placebo 3 (IQR 3 to 5). Consequently, the composite angina symptom score at pre-randomization was similar between both groups (*Table 1*).

For the primary outcome, data were available for all the 8631 follow-up days to unblinding. At 6-month follow-up, as compared to placebo, CTO PCI was more likely to improve the angina symptom score (OR 4.4, 95% CrI 1.6 to 12.7, Pr(Benefit)=0.996). This resulted from a clear reduction in the number of daily episodes of angina recorded on the ORBITA-app (OR 4.4 [95% CrI 1.6 to 11.8]; Pr(Benefit)=0.997) and a numerical reduction in daily use of anti-anginal medications (OR 1.5 [95% CrI 0.37 to 6.5]; Pr(Benefit)=0.715) (*Figure 2 and Table 3*).

Overall, patients undergoing CTO PCI, experienced on average over the 6 months of follow-up, 30.6 more days free of angina (95% CrI 11.1 to 50.7; Pr(Benefit)>0.999).

Patients allocated to CTO PCI had greater improvements in the SAQ domains of angina frequency (mean difference +10.7 [95% CrI 1.4 to 20.2]; Pr(Benefit)= 0.988), physical limitation (mean difference +13.5 [95% CrI 4.5 to 22.3]; Pr(Benefit) >0.999), and disease-specific quality of life (mean difference +18.2 [95% CrI 5.4 to 30.5]; Pr(Benefit)= 0.997). No clear between-group differences were observed for the angina stability or treatment satisfaction domains. The SAQ summary score improved more in the CTO PCI group as compared to placebo (mean difference +13.7 points [95% CrI 4.2 to 23.2]; Pr(Benefit)= 0.997; *Table 4*).

CTO PCI improved blinded physician assessed CCS class (-0.70 , 95% CrI -0.27 to -1.13 , $\text{Pr}(\text{benefit}) > 0.999$, *Figure 3*); however, there was little evidence of benefit in NYHA and Rose Dyspnea Scale grades. Similarly, there was no evidence of benefit of CTO PCI on EQ-5D-5L index values and visual analogue scale score (*Table 4*).

During the 6-month follow-up period, there were a total of 60 anti-anginal medication up-titrations (21 in the CTO PCI group and 39 in the placebo group). These occurred in 27 patients during follow-up (11 (44%) CTO PCI and 16 (64%) Placebo group). The distribution of anti-anginal medication classes used during follow-up was broadly similar between groups (Appendix, page 28).

In-hospital complications included 1 patient allocated to PCI who developed pericarditis and 1 patient allocated to placebo who had a pseudoaneurysm requiring thrombin injection. (Appendix, page 19).

Blinding was assessed at two timepoints. The primary assessment, performed after the randomized interventional procedure and overnight admission, just prior to discharge, gave a Bang blinding index of -0.08 (95% CI -0.19 to 0.02) in the CTO PCI group (all patients felt unable to guess treatment allocation) and 0.04 (95% CI -0.17 to 0.25) in the placebo group (seven patients felt able to guess treatment allocation, however three guessed incorrectly). The blinding index for ward and research staff was also close to zero in both groups, indicating no systematic ability to correctly identify treatment allocation (Appendix, p28).

Discussion

ORBITA-CTO is the first placebo-controlled trial to assess the efficacy of CTO PCI for relief of anginal symptoms. It confirms that CTO PCI was superior to a placebo procedure in improving the primary outcome of daily angina episodes as reported on the ORBITA-app, in patients with single vessel CTO subtending a viable and ischemic territory, and a complexity J-CTO score ≤ 3 . The results of ORBITA-CTO address a long-standing evidence gap in the management of symptomatic patients with CTO, providing placebo-controlled evidence that PCI is an effective treatment for anginal symptoms in this group of patients with complex stable coronary disease.

The symptomatic benefit observed in ORBITA-CTO was consistent across complementary measures, including daily angina frequency, blinded CCS class, and SAQ domains. The approximately 10-point between-group difference in SAQ angina frequency meets established criteria for moderate clinical importance, and the SAQ summary score increased by 13.7 points, substantially exceeding the 5-point threshold considered clinically meaningful. The magnitude of improvement was numerically greater than that reported in the unblinded EuroCTO trial but smaller than that observed in the open label COMET-CTO study.^{3,6} These differences likely reflect the impact of unblinded study designs on symptom reporting and treatment allocation crossover, and emphasize the value of placebo-controlled designs when evaluating symptom-based endpoints in complex PCI. Collectively, these data indicate a clinically meaningful reduction in angina burden with CTO PCI in appropriately selected patients.

Clinically, these findings support the option of offering CTO PCI to patients with angina attributable to a CTO, provided there is a high likelihood of technical success and acceptable procedural risk. The magnitude of benefit is moderate for angina, with no benefit seen in

physician assessed symptoms of dyspnoea, and a proportion of patients who did not achieve complete symptom resolution. Careful patient selection and shared decision-making therefore remain central to clinical application. Importantly, although in this trial anti-anginal therapy was withdrawn per protocol to isolate the physiological effect of CTO PCI, angina burden in the PCI group remained lower after revascularization than before intervention. While interactions between background antianginal therapy medication and revascularization cannot be excluded, it is unlikely that continuation of anti-anginal therapy would negate the symptomatic benefit observed. Consistent with this, model-based estimates showed that patients undergoing CTO PCI experienced approximately 31 additional angina-free days over 6 months compared with placebo.

Symptom improvement occurred in both groups; however, reduction in angina was immediate and sustained in the CTO PCI arm. Anti-anginal medical therapy was progressively up-titrated over follow-up, particularly in the placebo group, consistent with attenuation of placebo-related benefit. The overall treatment effect was observed despite inclusion of one unsuccessful PCI case reporting an average of 6 episodes of angina daily in the intention-to-treat analysis. This observation is consistent with prior evidence that unsuccessful CTO PCI is associated with poorer outcomes,¹⁸ and may also be associated with an increase in angina burden. The double-blind design of ORBITA-CTO allows quantification of the magnitude of placebo response associated with complex coronary intervention, which appears substantial and at least comparable to that observed in placebo-controlled trials of non-CTO PCI and coronary sinus reducer implantation.^{7,13,14} These findings reinforce the importance of placebo-controlled methodology when evaluating symptom-driven procedures.

Understanding the mechanisms underlying non-response is an important priority for future research. Potential explanations include non-ischemic drivers of symptoms, as is often the case in patients experiencing atypical angina or dyspnea, incomplete revascularization, and coronary microvascular dysfunction. Prior work has demonstrated that symptom patterns and their relationship to ischemia are heterogeneous, and these findings may extend to the CTO population.^{7,13,19} Conversely, the marked reduction in angina frequency observed in the placebo group after protocolized withdrawal of anti-anginal therapy might partly reflect hemodynamic effects, since collateral perfusion to CTO-dependent myocardium is pressure-gradient dependent and can be reduced by lower systemic arterial pressure or by vasodilator-induced redistribution of flow away from collateral-dependent territories.^{20,21} Planned subgroup analyses and dedicated coronary physiology sub-studies within this trial may help to identify clinical, anatomical, or physiological predictors of response, and thereby refine patient selection for CTO PCI.

Several strengths of ORBITA-CTO warrant emphasis. Primary among these was the use of a regional, multidisciplinary CTO and complex PCI MDT pathway that applied to all consecutive CTOs identified within the catchment area of the primary sites. This approach ensured thorough assessment of symptoms, ischemia, viability, and anatomical suitability weighed against anatomic complexity and patient-specific risk. Management decisions were made systematically rather than through referral to individual operators, thereby reducing selection bias inherent to studies recruiting for specialist procedures. Together, this framework enriched the study population for patients in whom angina could plausibly be attributed to the CTO, and in whom either optimized medical therapy, or CTO PCI could reasonably be expected to confer benefit. Such an approach represents a methodological benchmark for future CTO trials and a potential model for clinical practice when the aim is to

identify patients most likely to derive symptomatic benefit from complex invasive revascularization.

Procedural delivery was standardized and undertaken by experienced CTO operators using contemporary techniques, including hybrid crossing strategies and very high intravascular imaging use compared to contemporary registry data (92% vs 21%)¹⁰, resulting in high technical success and low complication rates. The technical success observed in this trial exceeds that reported in prior randomized CTO studies, such as EuroCTO and DECISION-CTO,^{5,6} and likely reflects both lesion selection (J-CTO complexity capped at 3) and the requirement for specialized dual-operator procedures.

The methodological strength of ORBITA-CTO arises from the robustness of blinding achieved in a setting where temporal masking is particularly challenging. CTO PCI procedures are typically prolonged and frequently require overnight admission, increasing the risk of inadvertent unblinding compared with same-day discharge in routine PCI. The ability to maintain masking in this context is central to the validity of the symptom findings. In trials of symptom-driven interventions, loss of blinding can exaggerate perceived benefit and undermine causal inference.²² The maintenance of blinding in ORBITA-CTO, together with the absence of crossover and consistent protocol delivery across two high-volume centers, strengthens confidence that the observed reduction in angina burden reflects a true treatment effect rather than expectancy or behavioral bias.

Several limitations should be acknowledged. Firstly, the study was conducted in two UK expert centers with highly experienced operators and mature CTO programs, which may limit generalizability to lower-volume centers or operators with less extensive experience. The use

of advanced techniques, high intravascular imaging utilization, and comprehensive periprocedural care may not be universally reproducible. Secondly, the sample size of 50 patients was determined at a time when the effect size and variance of the ORBITA-app were not known a priori. The trial was therefore not powered to a specific app-derived effect estimate. Instead, the design was informed by the expectation that high-frequency daily symptom capture, integrated with contemporaneous anti-anginal medication burden, would reduce measurement noise and improve signal-to-noise ratio compared with episodic questionnaire-based assessments such as the SAQ. This measurement strategy was anticipated to increase sensitivity to treatment effects and allow clinically meaningful symptomatic differences to be detected with a modest sample size. Consistent with this rationale, the observed separation between PCI and placebo was accompanied by credible intervals of clinically relevant magnitude, and subsequent studies within the ORBITA program have supported the responsiveness and reliability of this approach to symptom assessment.^{13,14} Thirdly, although complete withdrawal of anti-anginal therapy is not routine practice, post-revascularization adjustment of medical therapy is common and guided by symptoms. This study design prioritized internal validity while preserving clinical relevance to revascularization decisions in symptomatic patients on anti-anginal therapy. Finally, enrolment was restricted to lesions with moderate complexity and patients with the most complex or high-risk CTOs (J-CTO 4 & 5) excluded. These considerations are particularly relevant considering reports from less selective trials demonstrating higher complication rates, lower success, and potential harm signals when CTO PCI is performed outside of expert settings.²³

ORBITA-CTO provides the first placebo-controlled evidence that CTO PCI leads to a meaningful and sustained reduction in anginal symptoms. When performed in appropriately selected patients and in experienced centers, CTO PCI offers a valuable therapeutic option for

angina symptom relief beyond placebo. These findings should inform shared decision-making and future guideline discussions, while underscoring the importance of rigorous trial design in evaluating symptomatic benefits of complex coronary interventions.

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ORBITA-CTO Investigators

Sarosh Khan, Samer Fawaz, Rupert Simpson, Uzma Sajjad, Abdalla Ibrahim, Haroun Butt, Klio Konstantinou, Firas Al-Janabi, Paul Kelly, Reto Gamma, Rohan Jagathesan, Alamgir Kabir, Sean King, Debbie Sevant, Craig Robertson, Grigoris V Karamasis, Shah Mohdnazri, Kare Tang, Thomas R Keeble, John R Davies (Essex Cardiothoracic Centre, Mid and South Essex NHS Foundation Trust). Matthew Shun-Shin, Rasha Al-Lamee (Imperial College London). Peter O'Kane, Vivek Kadoth, Jonathan Hinton (University Hospitals of Dorset NHS Foundation Trust). James Spratt, Simon Wilson, Maciej Marciniak (St George's University Hospitals NHS Foundation Trust).

Contributors

SK, MSS, GVK, RKA-L, TRK and JRD were on the writing committee. SK, GVK, RKA-L, ESB, TRK and JRD were responsible for the conception and design of the study. SK and CR were responsible for symptom data acquisition. PK, SMN, KT, OMD, CC were responsible for angiography data assessment and collection. JRD, PDO'K, JCS were principal investigators at individual trial sites. JRD, KT, SMN and PDO'K were responsible for carrying out the invasive placebo and coronary interventional procedures. SF, RS, US, AI, HB were responsible for invasive data collection and randomization. MSS was the study statistician. SK, MSS, RKA-L, TRK and JRD were responsible for the data interpretation and writing of the report. SK, MSS and JRD directly assessed and verified the data reported in this manuscript. All authors were responsible for reviewing the manuscript. All authors had access to all the included data. SK and JRD were responsible for the decision to submit for publication.

Data sharing

The study protocol and statistical analysis plan are available in the appendix. Individual patient data will not be made available.

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Perspectives

Clinical Competencies

Competency in Medical Knowledge:

In patients with symptomatic single-vessel chronic total occlusion and objective evidence of ischemia and viability, CTO PCI results in a reduction in angina burden beyond placebo, confirming a true symptomatic benefit attributable to revascularization rather than expectation alone.

Competency in Systems-Based Practice:

Given the procedural complexity, resource intensity, and potential for placebo effects, CTO PCI should be performed within experienced centers using structured multidisciplinary team assessment to ensure appropriate utilization of health-care resources.

Translational Outlook

Translational Outlook 1:

Future studies should explore how physiology, imaging, and symptom-based markers can be integrated to refine patient selection and identify those most likely to benefit from CTO PCI.

Translational Outlook 2:

Longer-term follow-up and health-economic analyses are needed to determine whether the symptomatic improvements observed with CTO PCI translate into sustained quality-of-life gains and reduced health-care utilization.

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Figure Legends

Figure 1. Trial Profile

Figure 2 Primary Outcomes

Individual patient data, from pre-randomization to the end of the blinded follow up period, for A) The primary efficacy endpoint of angina symptom score. B) Daily angina episodes as reported via the ORBITA-app. C) Antianginal medication units

Figure 3 Distribution of CCS Angina Severity Class

Shows the distribution of the physician-assessed Canadian Cardiovascular Society (CCS) angina severity class prior to randomization and at the end of follow-up prior to unblinding. The CCS class ranges from 0 to IV, with class 0 indicating no angina and class IV indicating angina at rest.

Central Illustration. The ORBITA-CTO Trial.

Table 1 Pre-randomization characteristics of study patients

Data are median (IQR) or n (%). Left ventricular systolic function was defined as normal ($\geq 55\%$), mildly impaired (45–54%), or moderately impaired (35–44%). CTO PCI=chronic total occlusion percutaneous coronary intervention. CABG=coronary artery bypass grafting. J-CTO = Japan CTO score. PROGRESS CTO= Prospective Global Registry for the Study of Chronic Total Occlusion Intervention score. J-CTO and PROGRESS scores performed on randomization procedure after dual injection coronary angiograms.

	CTO PCI group (n=25)	Placebo group (n=25)
Age		
Mean $\pm$ SD	62 $\pm$ 8.7	64 $\pm$ 11.9
Median (IQR)	61 (57-69)	67 (56-71)
Sex		
Male	18 (72%)	19 (76%)
Female	7 (28%)	6 (24%)
Ethnicity		
White	24 (96%)	24 (96%)
Asian	1 (4%)	1 (4%)
Hypertension	13 (52%)	16 (64%)
Diabetes		
Non-insulin dependent	4 (16%)	6 (24%)
Insulin-dependent	0	0
Hyperlipidemia	24 (96%)	24 (96%)

Prev CABG	2 (8%)	1 (4%)
Previous PCI	17 (68%)	12 (48%)
Smoking status		
Never smoked	8 (32%)	9 (36%)
Ex-smoker	12 (48%)	12 (48%)
Current smoker	5 (20%)	4 (16%)
Left ventricular systolic function		
Normal	23 (92%)	21 (84%)
Mild impairment	1 (4%)	0 (0%)
Moderate impairment	1 (4%)	4 (16%)
Median reversible ischemia burden	15% (13-18)	19% (12-24)
CTO vessel		
RCA	17 (68%)	13 (52%)
LAD	6 (24%)	10 (40%)
LCX	2 (8%)	2 (8%)
In-stent CTO	3 (12%)	3 (12%)
J-CTO score		
0	6 (24%)	8 (32%)
1	4 (16%)	6 (24%)
2	12 (48%)	5 (20%)
3	3 (12%)	6 (24%)
J-CTO score		

Mean	1.48 ± 1.00	1.36 ± 1.19
Median	2 (IQR 1 to 2)	1 (IQR 0 to 2)
PROGRESS score		
0	9 (36%)	10 (40%)
1	12 (48%)	11 (44%)
2	3 (12%)	3 (12%)
3	1 (4%)	1 (4%)
PROGRESS score		
Mean	0.84 ± 0.80	0.80 ± 0.82
Median	1 (IQR 0 to 1)	1 (IQR 0 to 1)
Canadian Cardiovascular Society Class		
0	0	1 (4%)
I	1 (4%)	1 (4%)
II	16 (64%)	17 (68%)
III	8 (32%)	6 (24%)
Rose Dyspnea Scale		
0	3 (12%)	2 (8%)
1	10 (40%)	12 (48%)
2	10 (40%)	8 (32%)
3	1 (4%)	2 (8%)
4	1 (4%)	1 (4%)
New York Heart Association Class		

I	5 (20%)	3 (12%)
II	17 (68%)	16 (64%)
III	3 (12%)	6 (24%)
EQ-5D-5L		
EQ-5D index value		
Mean	0.64 ± 0.19	0.66 ± 0.23
Median	0.65 (IQR 0.59 to 0.75)	0.65 (IQR 0.57 to 0.84)
EuroQol visual analogue scale		
Mean	64.2 ± 17.8	66.4 ± 18.7
Median	70.0 (IQR 55.0 to 75.0)	66.0 (IQR 60.0 to 70.0)
Seattle Angina Questionnaire		
Angina frequency domain		
Mean	59.6 ± 17.2	57.6 ± 15.1
Median	60.0 (IQR 50.0 to 70.0)	60.0 (IQR 50.0 to 70.0)
Physical limitation domain		
Mean	60.4 ± 21.2	55.4 ± 21.9
Median	63.9 (IQR 47.2 to 77.8)	58.3 (IQR 38.9 to 72.2)
Angina stability		
Mean	50.0 ± 14.4	48.0 ± 24.9
Median	50.0 (IQR 50.0 to 50.0)	50.0 (IQR 25.0 to 50.0)
Quality of life domain		
Mean	38.0 ± 18.0	34.7 ± 18.9
Median	41.7 (IQR 25.0 to 50.0)	33.3 (IQR 25.0 to 41.7)
Treatment satisfaction		

Mean	86 ± 17.8	88.0 ± 14.2
Median	93.75 (IQR 81.3 to 100)	93.5 (IQR 75.0 to 100)
SAQ summary score		
Mean	52.7 ± 14.3	49.2 ± 15.3
Median	57.0 (IQR 43.3 to 64.8)	52.0 (IQR 37.0 to 59.1)
Median angina duration prior to enrolment, months	12 (8-18)	12 (8-24)
Daily angina frequency		
Mean	1.3 ± 1.2	1.5 ± 1.2
Median	0.93 (IQR 0.57 to 1.29)	1.5 (IQR 0.78 to 1.78)
Number of antianginal units		
Mean	4.44 ± 1.45	4.00 ± 1.53
Median	4 (IQR 3 to 5)	3 (IQR 3 to 5)
Angina symptom score		
Mean	32.4 ± 10.2	29.5 ± 10.7
Median	29.3 (IQR 22.2 to 37.6)	26.2 (IQR 22.5 to 35.8)

Table 2 Procedural characteristics

Randomization procedural characteristics of both groups. One patient from CTO PCI group had unsuccessful CTO crossing. CTO PCI= chronic total occlusion percutaneous coronary intervention.

	PCI (n=25)	Placebo (n=25)
Dual access		
Bi-radial	16 (67%)	14 (56%)
Radial-femoral	8 (29%)	10 (40%)
Bi-femoral	1 (4%)	1 (4%)
Successful crossing technique		
AWE	17 (68%)	-
RWE	5 (20%)	-
RDR	2 (8%)	-
Unsuccessful crossing	1 (4%)	-
Technical success	24 (96%)	-
Intravascular ultrasound guided procedure	23 (92%)	-
Mean total length of stents used, mm	69.4 ± 29.8	-
Median largest stent diameter, mm	3.5 (IQR 3 to 3.5)	-
Median number of stents used	2 (IQR 1 to 3)	-
Procedure duration, mins	191 ± 52	104 ± 14
Mean fluoroscopy time, min	45 ± 29	5 ± 3
Mean contrast volume, mls	255 ± 87	86 ± 47
Mean patient dose, mGy	1861 ± 1205	184 ± 138

Mean dose Area Product (DAP), Gy.cm ²	147 ± 101	21 ± 19
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Table 3 Primary symptom ORBITA-app outcomes at end of follow-up

Data are odds ratio (95% credible interval) and probability of benefit. CTO PCI=chronic total occlusion percutaneous coronary intervention.

	Odds ratio [95% CrI] CTO PCI vs placebo	Pr(Benefit) with CTO PCI vs placebo
Angina symptom score	4.38 [1.57 to 12.69]	0.996
Daily episodes of angina	4.38 [1.55 to 11.78]	0.997
Standardized units of anti-anginal	1.54 [0.37 to 6.46]	0.715

Table 4 Secondary endpoints

Data are score values (95% credible interval) and probability benefit. The follow-up and increment values are model based estimates (to avoid floor and ceiling effects), for an exemplar patient, conditional on the median pre-randomization value. CTO PCI=chronic total occlusion percutaneous coronary intervention. SAQ=Seattle Angina Questionnaire.

	n	Score increment	Score at 6-month follow-up	Benefit from pre-randomization to follow-up	Probability of benefit with CTO PCI vs placebo
SAQ angina frequency (pre-randomization median 60)					
CTO PCI	24	26.6 (18.1 to 33.8)	86.6(78.1 to 93.8)	10.7	0.988
Placebo	25	15.8 (6.5 to 25.2)	75.8 (66.5 to 85.2)	(1.4 to 20.2)	
SAQ physical limitation (pre-randomization median 59.7)					
CTO PCI	24	27.4 (18.7 to 34.0)	87.1 (79.5 to 93.7)	13.5	>0.999
Placebo	25	13.7 (3.7 to 23.1)	73.4 (63.4 to 82.9)	(4.5 to 22.3)	
SAQ angina stability (pre-randomization median 50.0)					
CTO PCI	24	7.7 (-0.8 to 17.3)	57.7 (49.2 to 67.3)	6.6	0.898
Placebo	25	1.2 (-9.0 to 11.9)	51.2 (41.0 to 61.9)	(-3.7 to 17.4)	
SAQ quality of life (pre-randomization median 41.7)					
CTO PCI	24	27.4 (16.7 to 37.6)	69.1 (58.3 to 79.2)	18.2	0.997
Placebo	25	9.1 (-3.0 to 21.0)	50.8 (38.7 to 62.7)	(5.4 to 30.5)	
SAQ treatment satisfaction (pre-randomization median 93.8)					

CTO PCI	24	1.2 (-2.9 to 4.3)	95.0 (90.8 to 98.1)	4.5	0.971
Placebo	25	-3.3 (-8.3 to 0.9)	90.4 (85.5 to 94.7)	(-0.3 to 9.2)	
SAQ summary score (pre-randomization median 52.9)					
CTO PCI	24	25.8 (17.3 to 33.1)	78.7 (70.2 to 86.0)	13.7	0.997
Placebo	25	12.0 (1.9 to 21.7)	64.8 (54.8 to 74.6)	(4.2 to 23.2)	
Canadian Cardiovascular Society Class (pre-randomization median 2.0)					
CTO PCI	24	-1.6 (-1.8 to -1.3)	0.4 (0.2 to 0.7)	-0.7	>0.999
Placebo	25	-0.9 (-1.3 to -0.5)	1.1 (0.7 to 1.5)	(-1.1 to -0.3)	
Rose Dyspnea Scale grade (pre-randomization median 2.0)					
CTO PCI	24	-1.1 (-1.5 to -0.8)	0.9 (0.5 to 1.2)	-0.3	0.948
Placebo	25	-0.8 (-1.2 to -0.4)	1.2 (0.8 to 1.6)	(-0.8 to 0.1)	
New York Heart Association class (pre-randomization median 2.0)					
CTO PCI	24	-0.6 (-0.8 to -0.4)	1.4 (1.2 to 1.6)	-0.1	0.812
Placebo	25	-0.5 (-0.7 to -0.3)	1.5 (1.3 to 1.7)	(-0.4 to 0.1)	
EQ-5D index value (pre-randomization median 0.7)					
CTO PCI	24	0.1 (0.03 to 0.2)	0.75 (0.68 to 0.82)	0.05	0.895
Placebo	25	0.1 (-0.03 to 0.1)	0.70 (0.63 to 0.78)	(-0.03 to 0.13)	
EuroQol visual analogue scale (pre-randomization median 66.0)					
CTO PCI	24	8.9 (1.3 to 15.0)	74.9 (67.3 to 81.0)	4.1	0.847
Placebo	25	4.7 (-3.3 to 11.9)	70.7 (62.7 to 77.9)	(-3.7 to 12.3)	

306 did not meet inclusion criteria

29 excluded

23 non revascularized non-CTO vessel
2 ischaemia in non-CTO vessel
1 exclusion to PCI or DES
1 intolerant to DAPT
1 life expectancy <2 years
1 inability to consent

83 eligible for
enrolment

27 declined to participate

56 enrolled

6 excluded after enrolment

5 recanalized CTO
1 asymptomatic

3 temporarily withdrawn after enrolment and
later re-enrolled

2 bystander disease requiring PCI
(both exited and re-enrolled)
1 Upper GI bleed pre-randomisation
(exited and re-enrolled after treatment)

50 randomly
assigned

25 allocated to CTO PCI

25 allocated to placebo intervention

25 underwent CTO PCI
1 unsuccessful CTO PCI
1 complication leading to
withdrawal

25 available for intention-to-treat
analysis of primary outcome

25 available for intention-to-treat
analysis of primary outcome

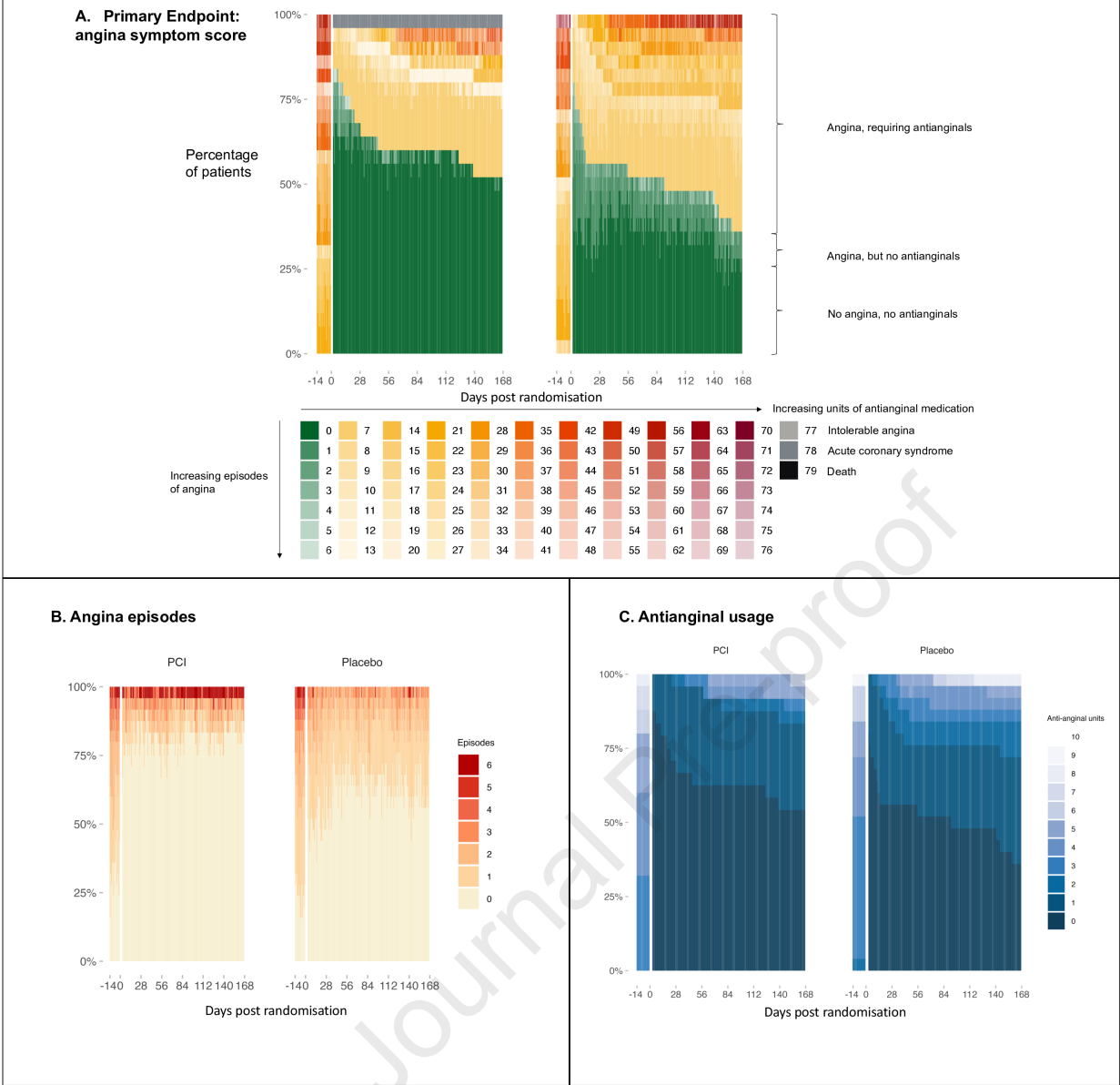


Figure 2: Primary outcome

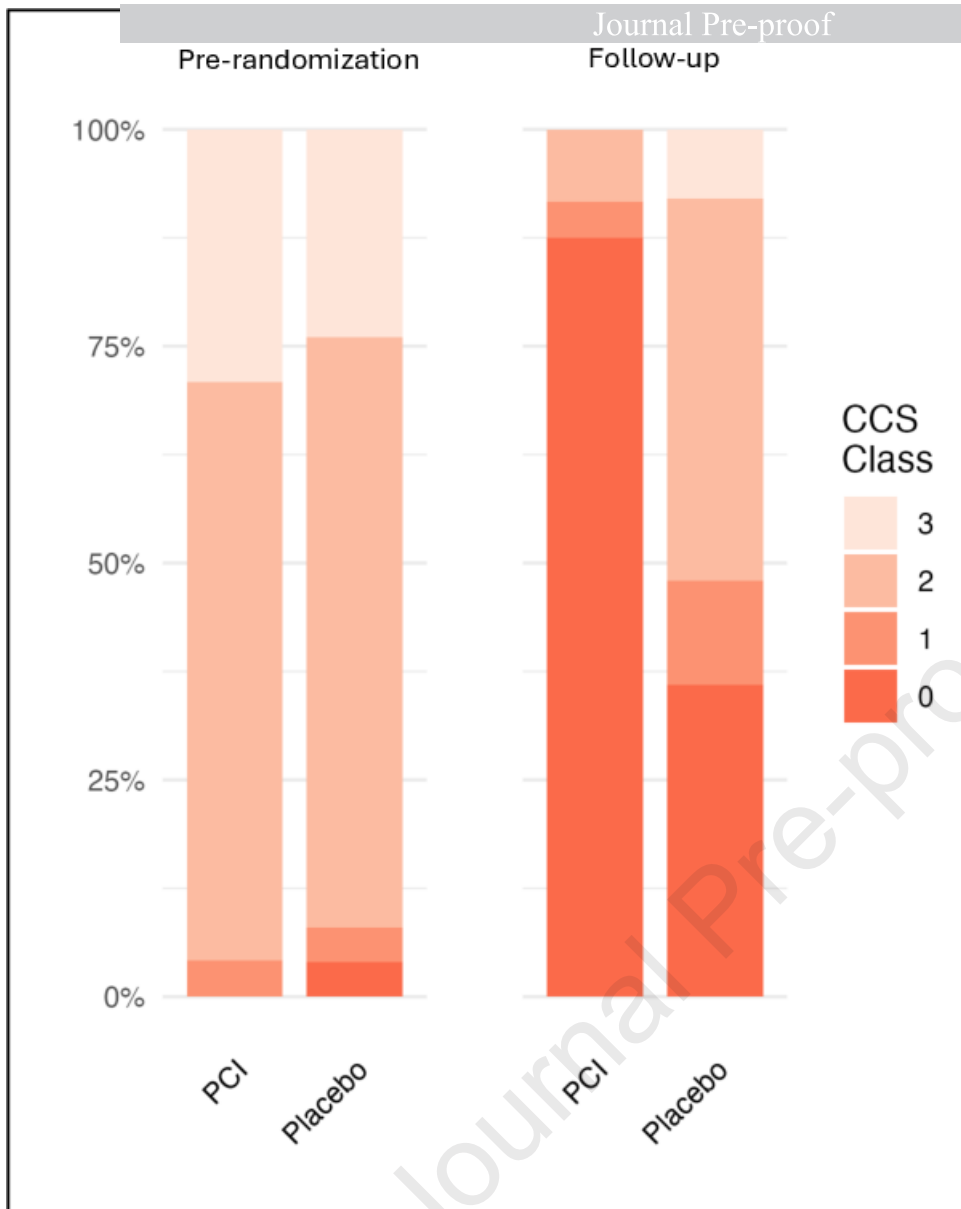


Figure 3: Secondary outcome CCS class

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CENTRAL ILLUSTRATION: A Randomized, Placebo-Controlled Trial of Chronic Total Occlusion Percutaneous Coronary Intervention in Stable Angina—ORBITA-CTO

This is the first randomized, double-blind, placebo-controlled trial evaluating CTO PCI, addressing a major evidence gap in a field where procedural risks are higher and prior trials have been unblinded. It provides the first data isolating the treatment effect of CTO PCI from the substantial placebo response known to accompany revascularization procedures.



In patients with symptomatic single-vessel chronic total occlusion (CTO), percutaneous coronary intervention (PCI) **produced a measurable reduction in angina burden** compared with a placebo procedure.

TRIAL DESIGN

- Investigator-initiated
- Multicenter
- Double-blind
- Randomized 1:1 to CTO PCI or placebo procedure
- Blinded 24-week follow-up

POPULATION

50 patients with

- symptomatic, single-vessel CTO (J-CTO ≤ 3)
- evidence of ischemia and viability
- no significant non-CTO coronary disease
- optimized anti-anginal therapy before randomization

64

median age

74%

men

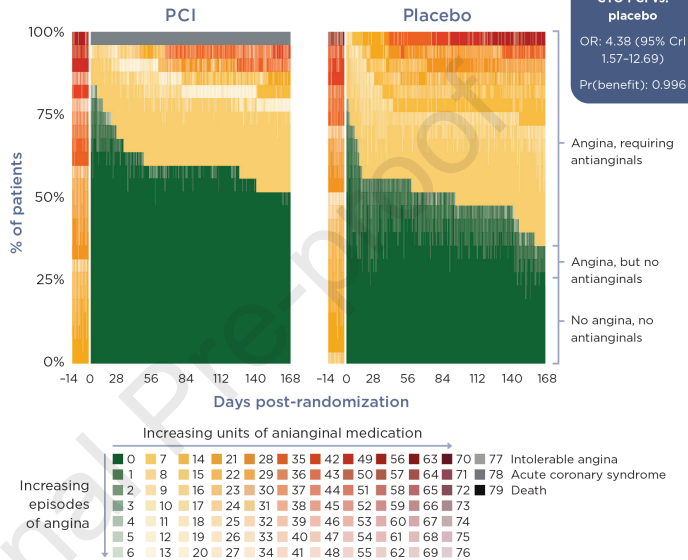
94%

CCS class II–III at baseline

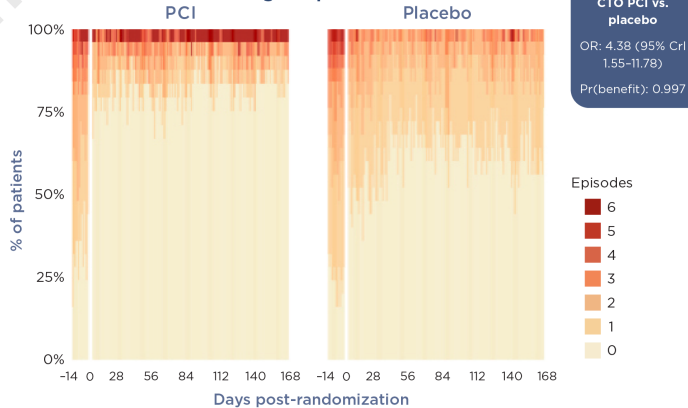


RESULTS

Primary endpoint: Angina symptom score



Angina episodes



Over 6 months, patients undergoing CTO PCI experienced **~31 additional angina-free days** compared with placebo.

Distribution of CCS angina severity class

